ANAT 1009 – Lab #4: The Nervous System

Purpose:

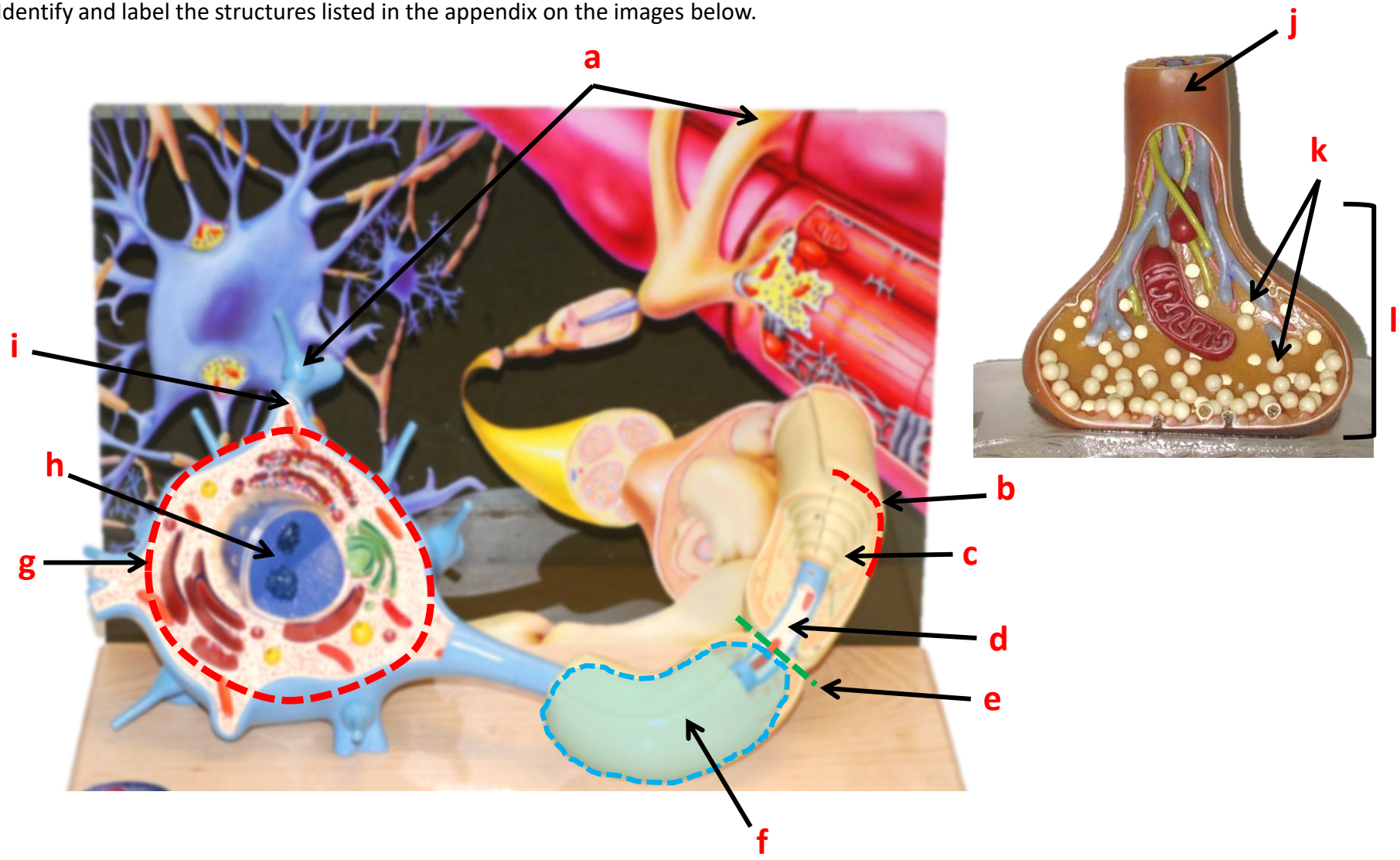
- To identify the structures of the nervous system

Video Available:
Nervous System
Microanatomy

The Appendix, located at the end of the lab, lists the structures for which you are responsible.

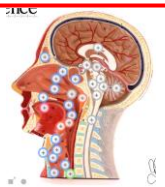
Station 1: The Neuron and Synapse

1. Identify and label the structures listed in the appendix on the images below.



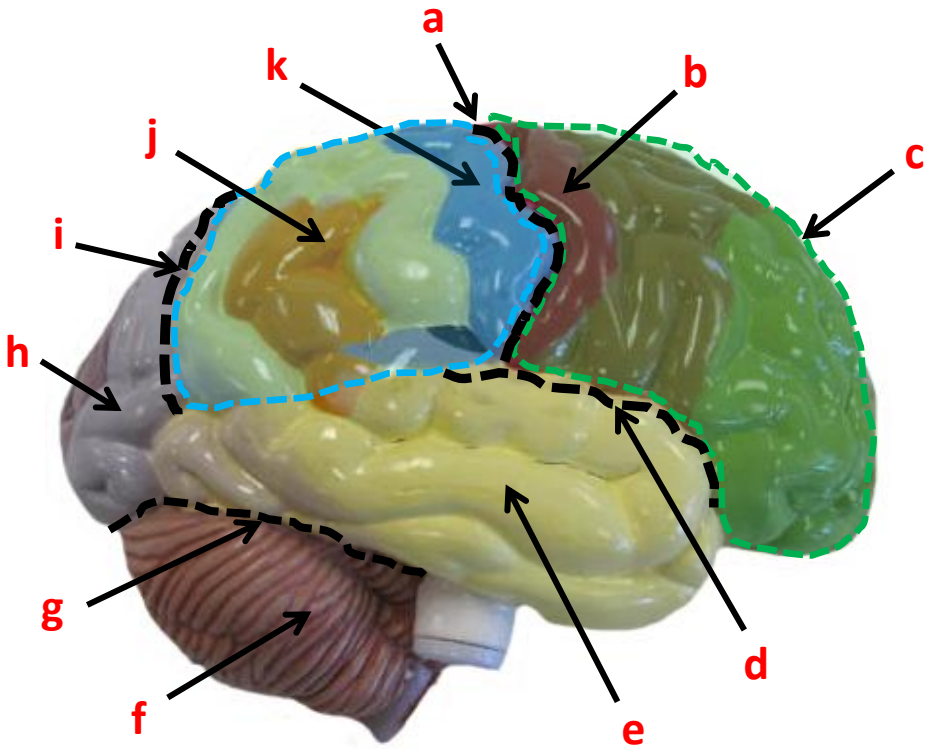
Station 2: The Brain

1. Identify and label the structures listed in the appendix on the images below.

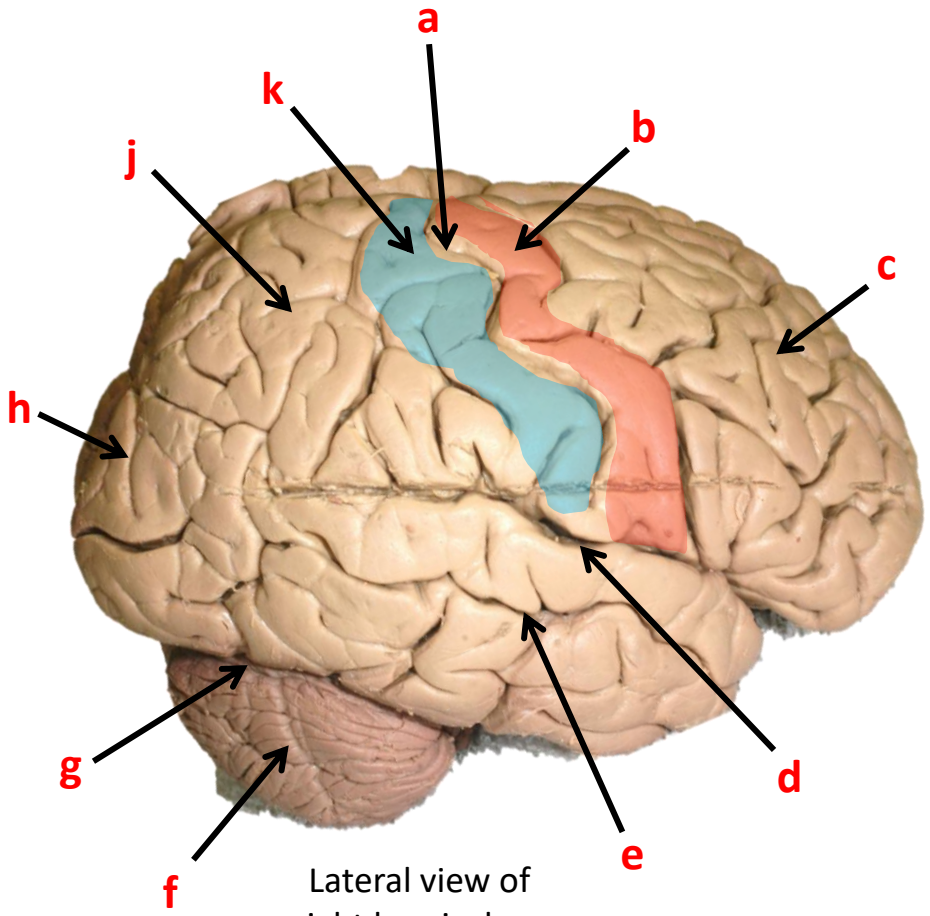


Rotational Image Available:
Rotational Images -> Nervous System -> Sagittal Head and Neck

Video Available:
Central Nervous System Introduction



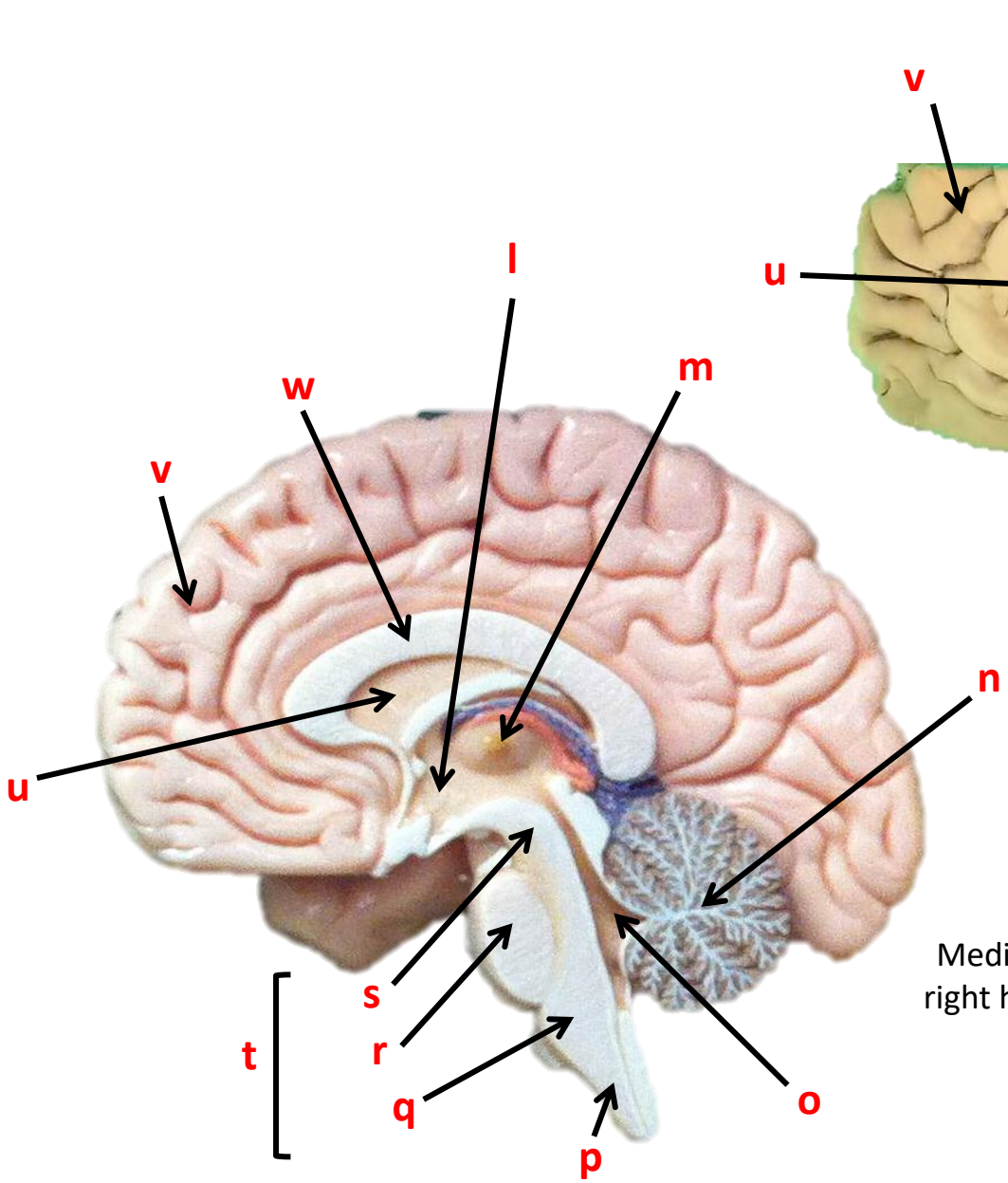
Lateral view of right hemisphere



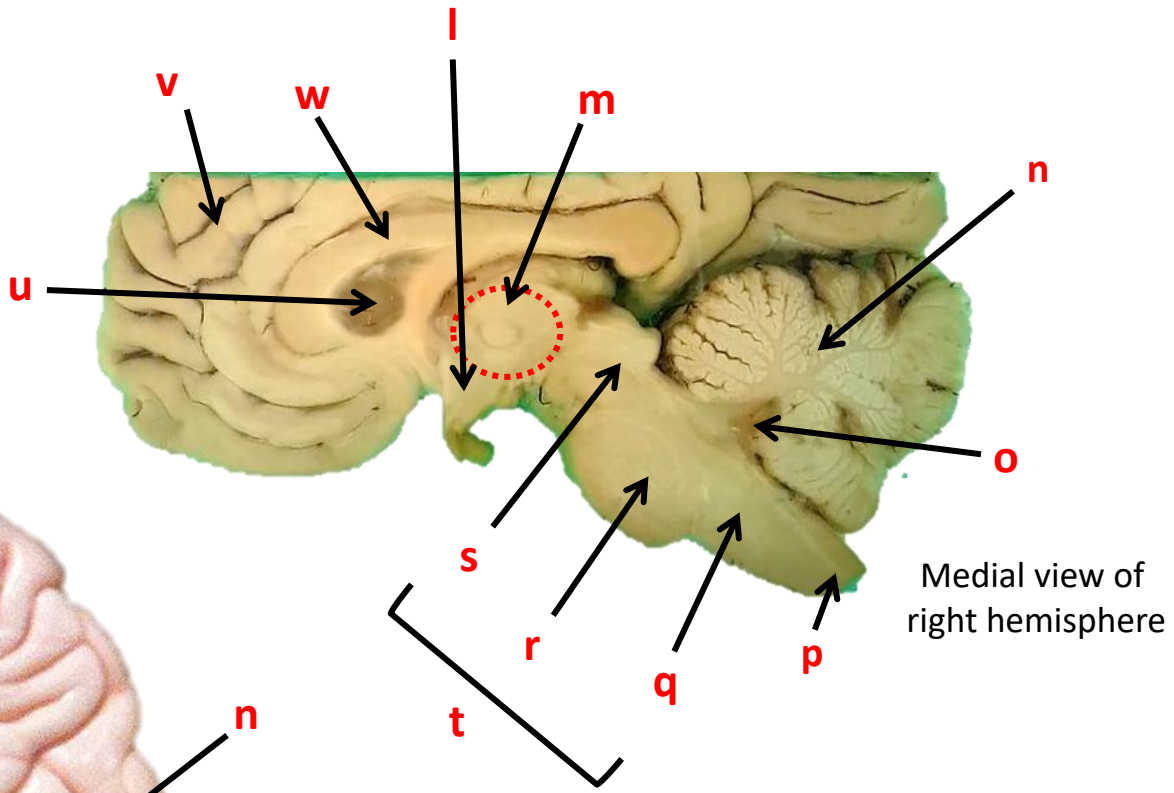
Lateral view of right hemisphere

Station 2: The Brain (continued)

1. Identify and label the structures listed in the appendix on the images below.



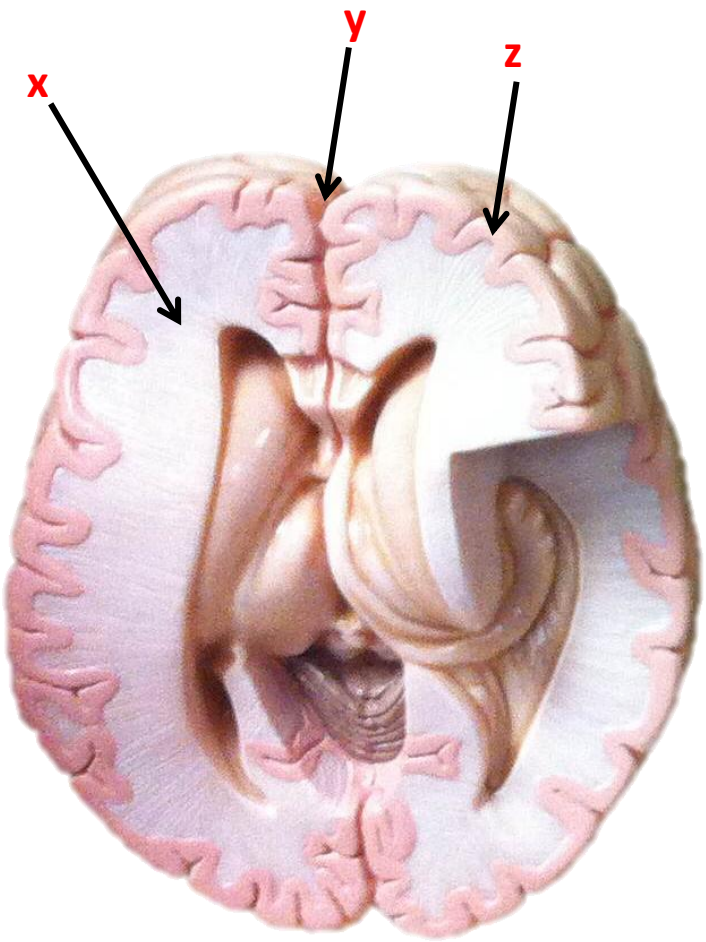
Medial view of right hemisphere



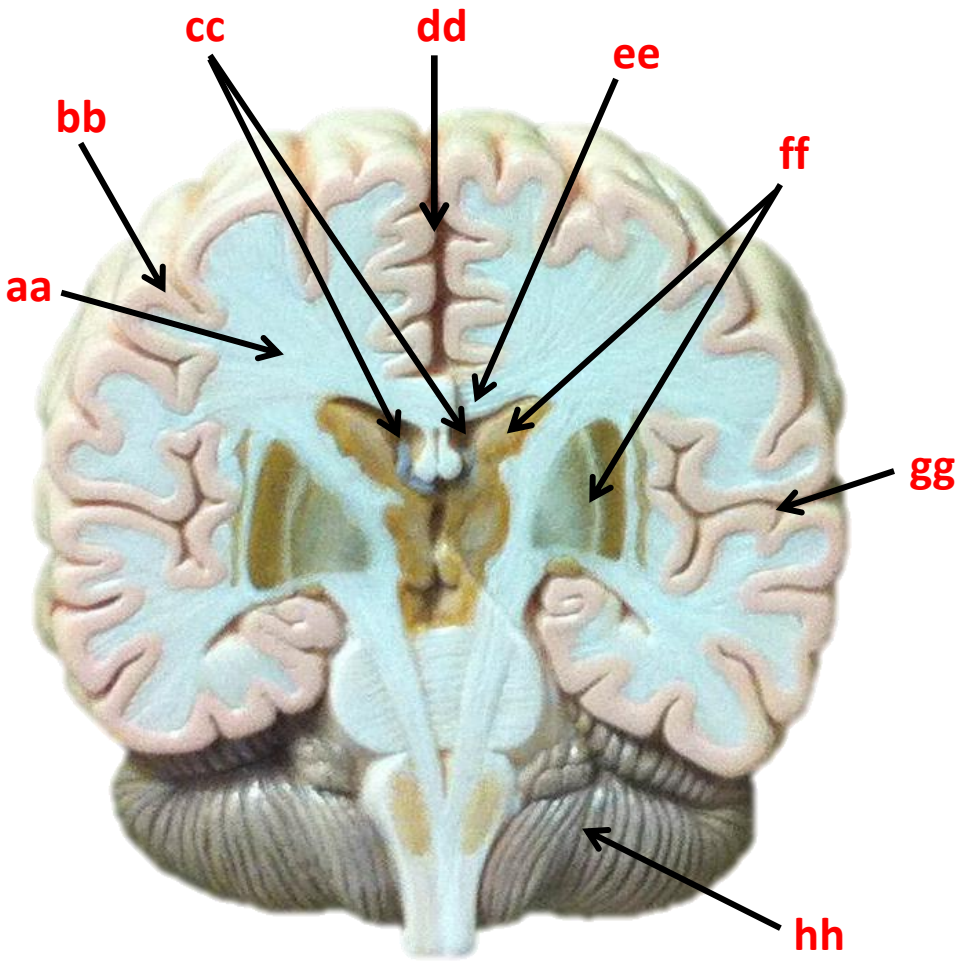
Medial view of right hemisphere

Station 2: The Brain (continued)

1. Identify and label the structures listed in the appendix on the images below.



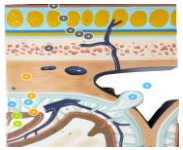
Superior view of
transverse section



Anterior view of
coronal/frontal section

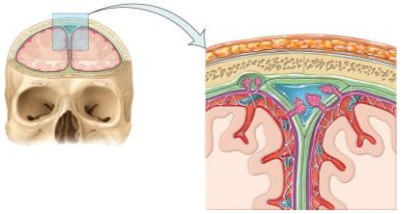
Station 3: Cerebrospinal Fluid (CSF) Flow

1. Identify and label the structures listed in the appendix on the images below.

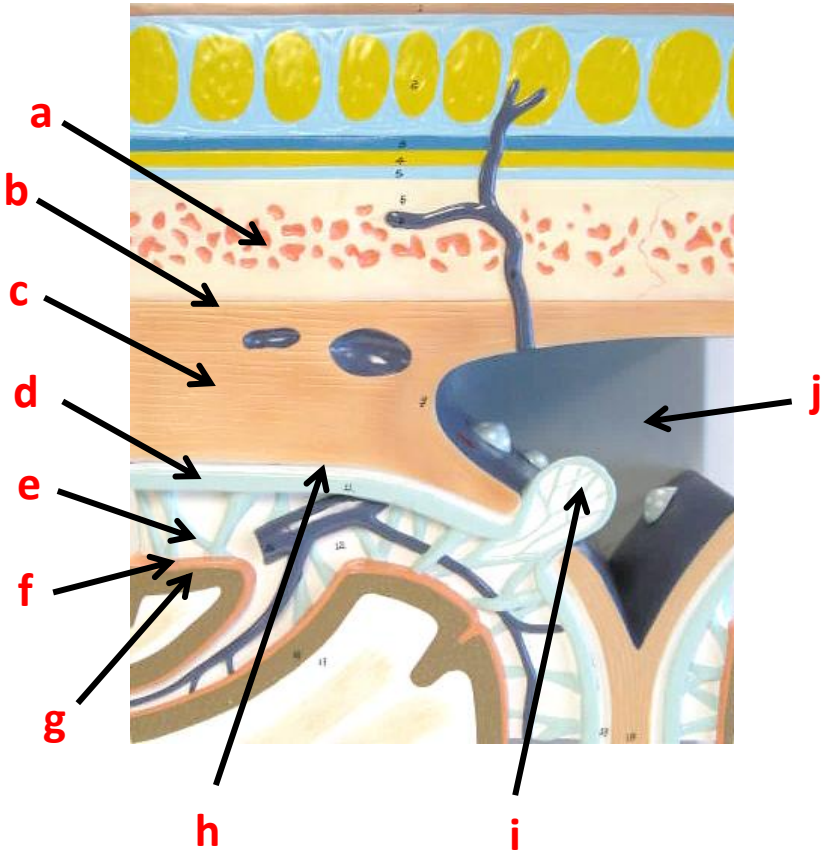


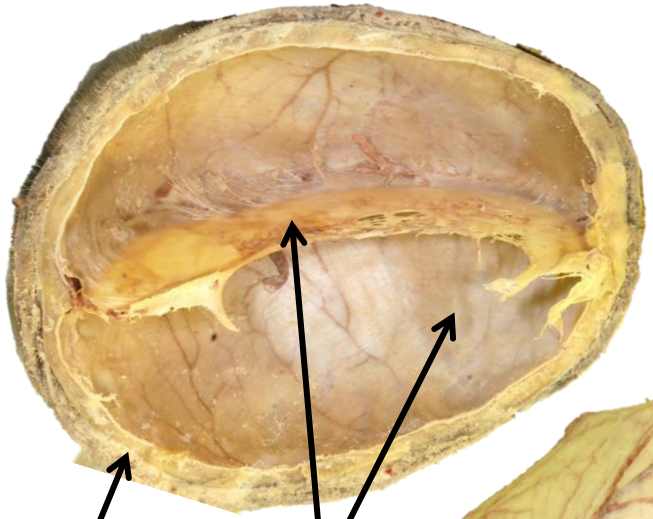
Rotational Image Available:
Rotational Images ->
Nervous System ->
Meninges

Video Available:
Nervous System
Microanatomy

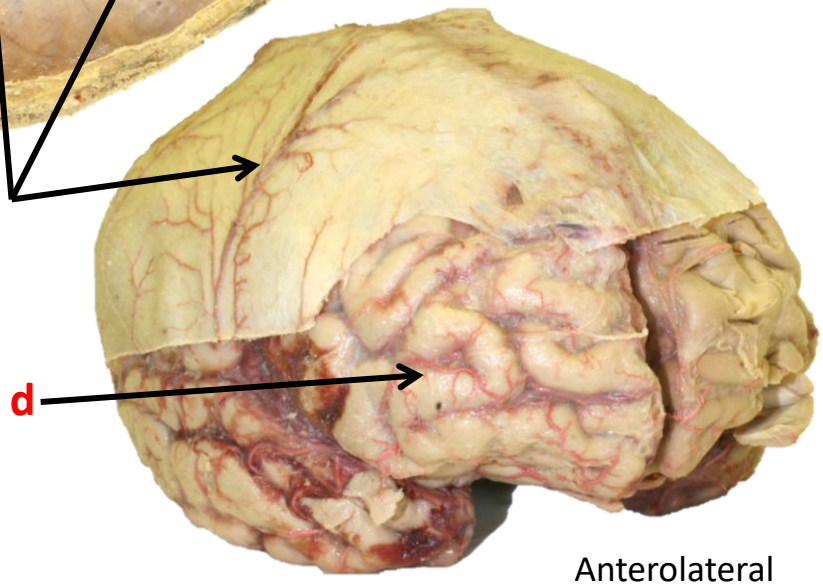


Frontal section through the scalp, skull, meninges and brain

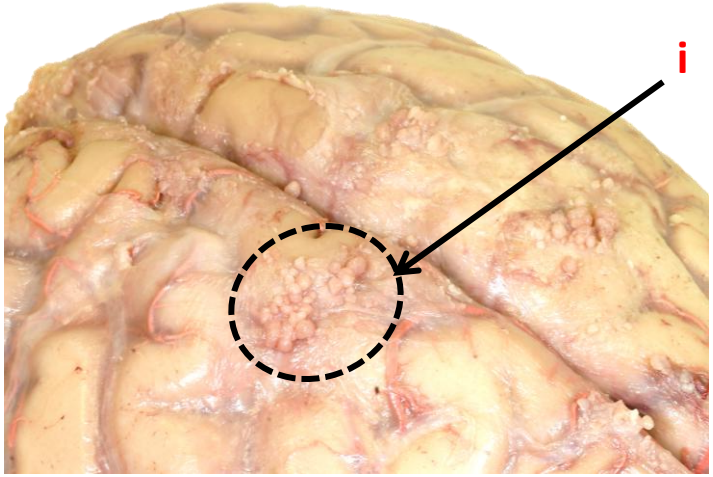




Inferior view of top portion of skull



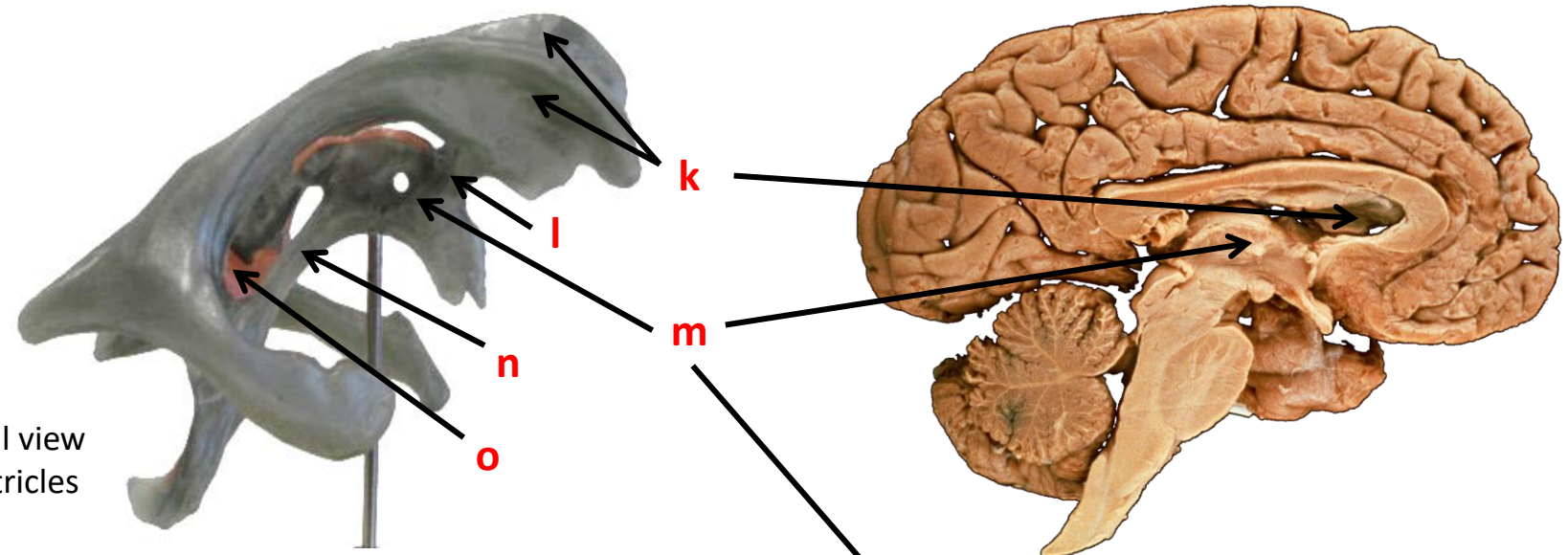
Anterolateral view of brain



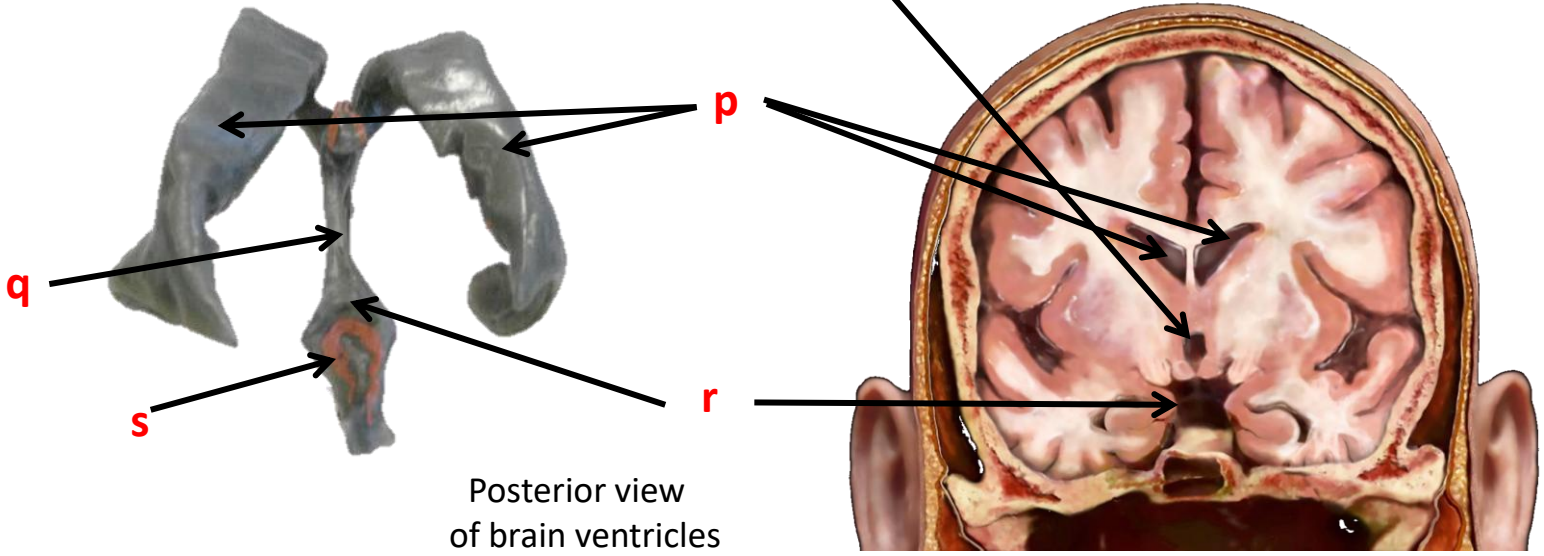
Superior view of brain and arachnoid

Station 3: Cerebrospinal Fluid (CSF) Flow – Ventricular System

1. Identify and label the structures listed in the appendix on the images below.



Anterolateral view
of brain ventricles



Posterior view
of brain ventricles

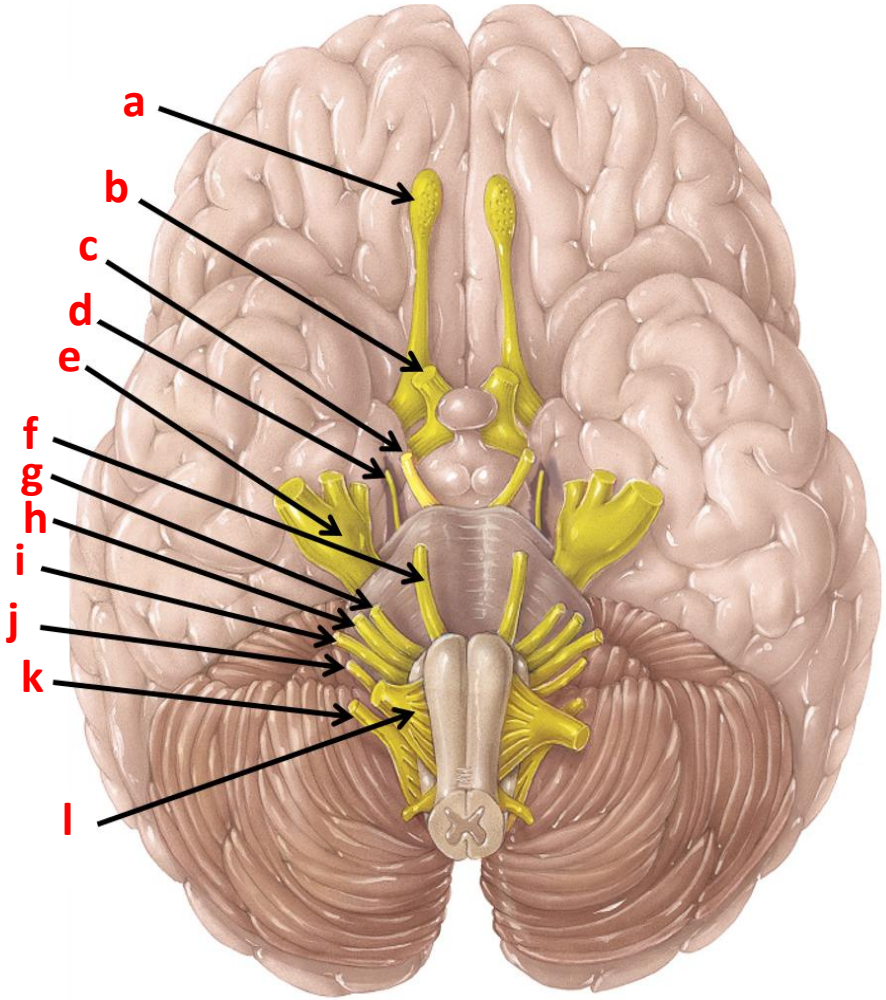
Questions:

1. What is the purpose of CSF?

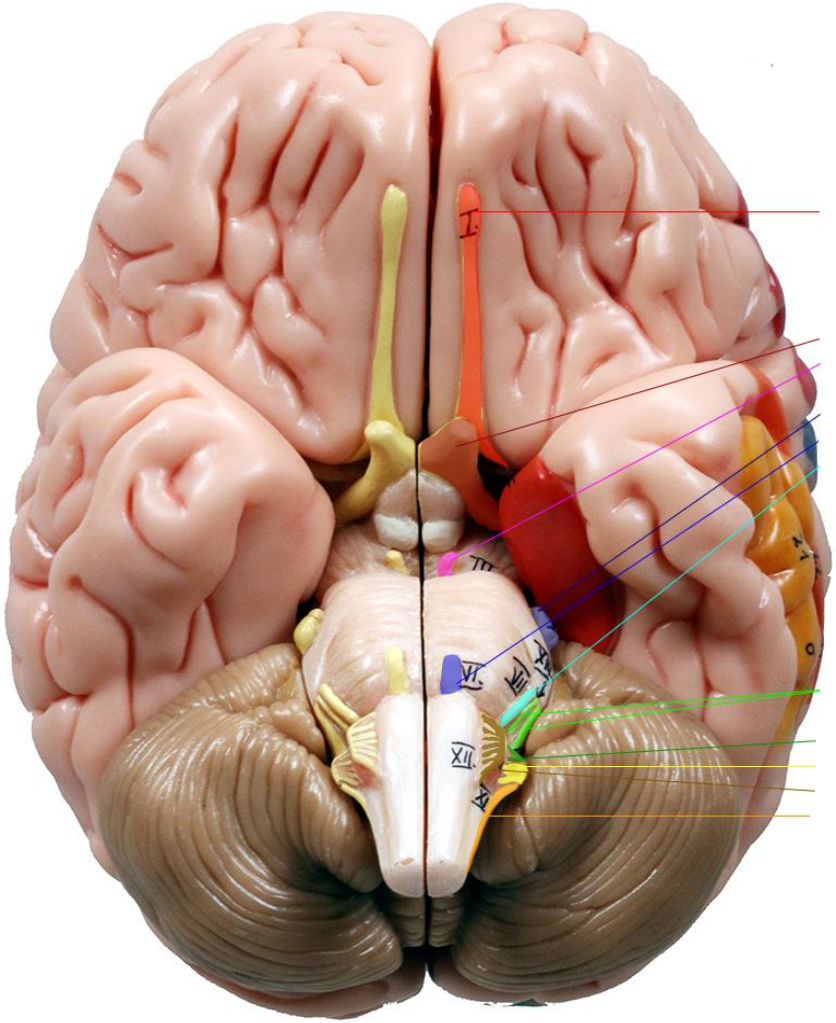
2. In your own words, describe the relationship of the three meningeal layers and their functions.

Station 4: Cranial Nerves

1. Identify and label each of the cranial nerves on the images below.



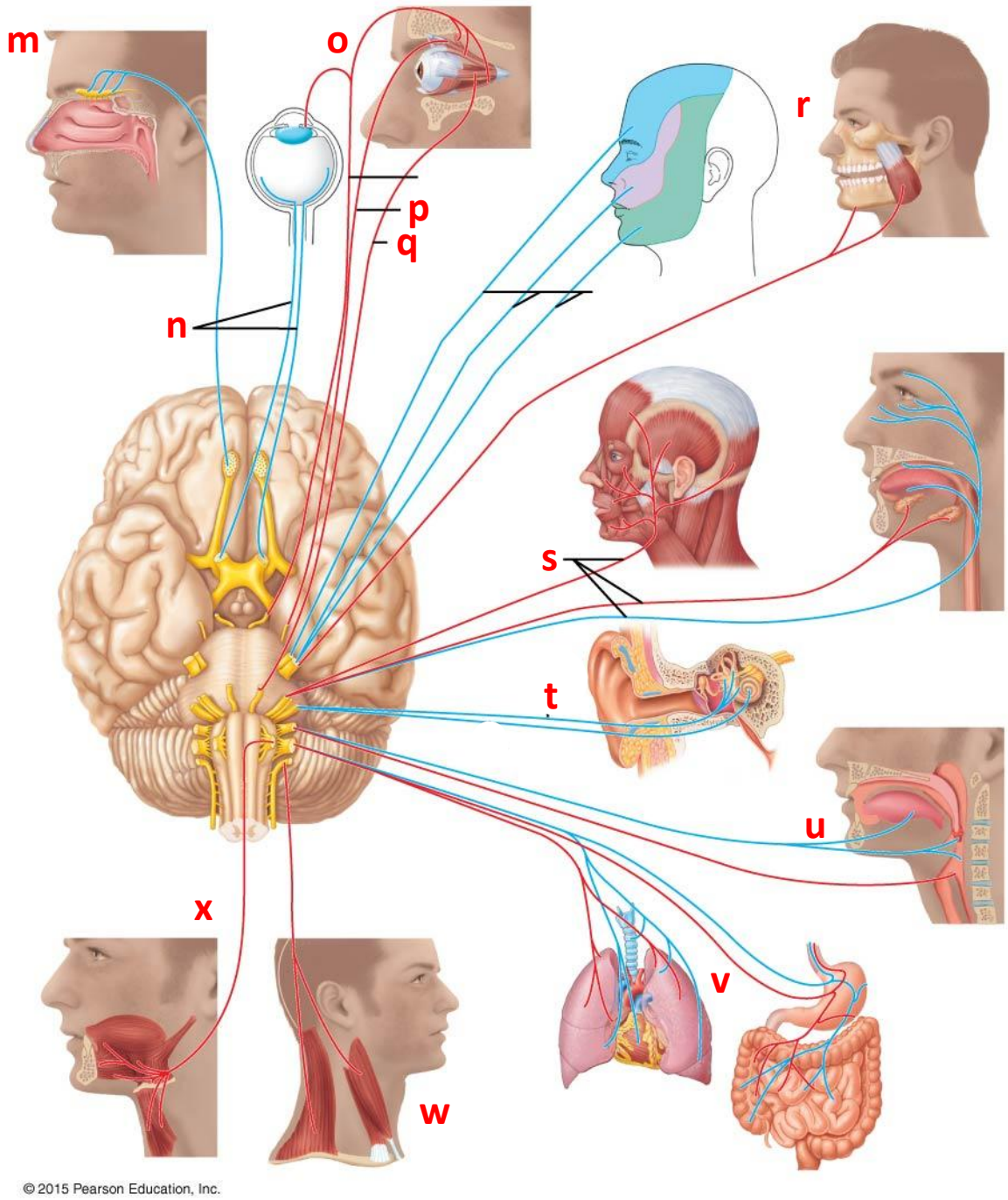
Inferior view of the brain and brainstem



Inferior view of the brain and brainstem

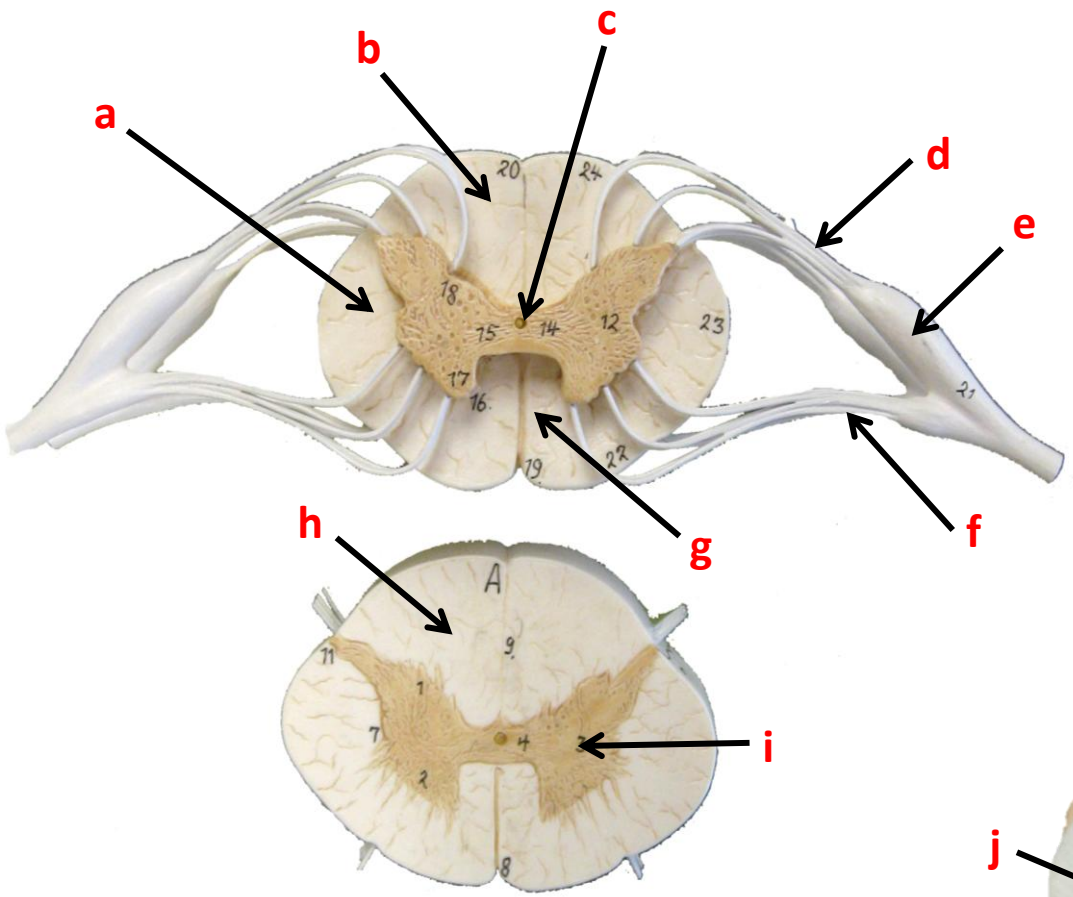
Station 4: Cranial Nerves (continued)

1. Identify and label each of the cranial nerves on the images below. Refer to your textbook to help you with the functions of each cranial nerve.

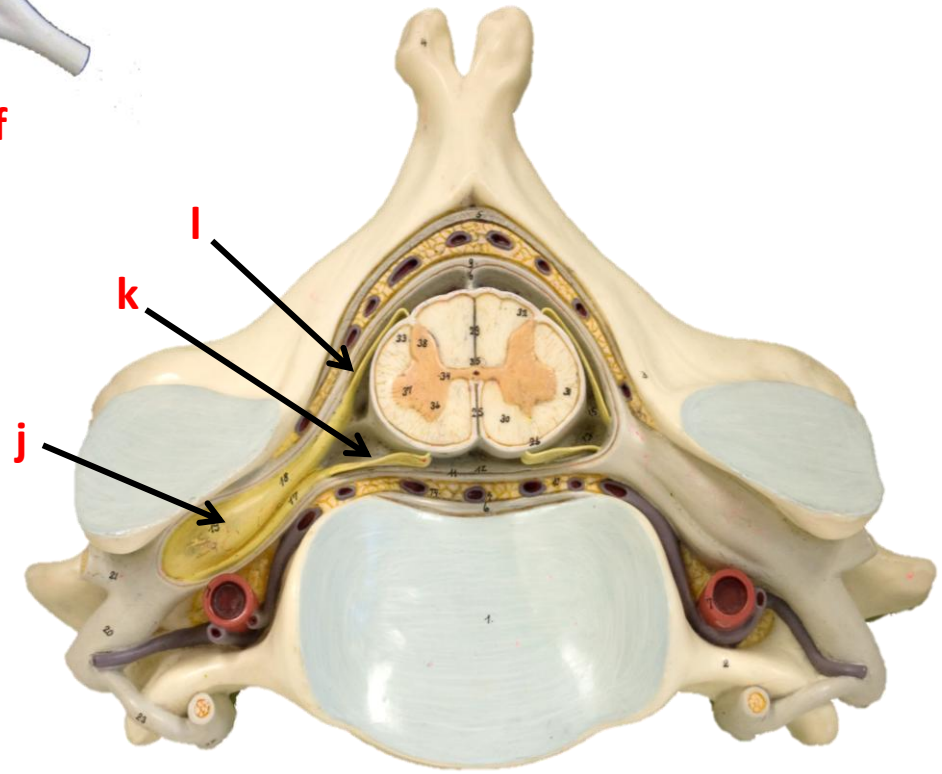


Station 5: The Spinal Cord

1. Locate the main structures of the spinal cord listed in the appendix, and record them on the images below.



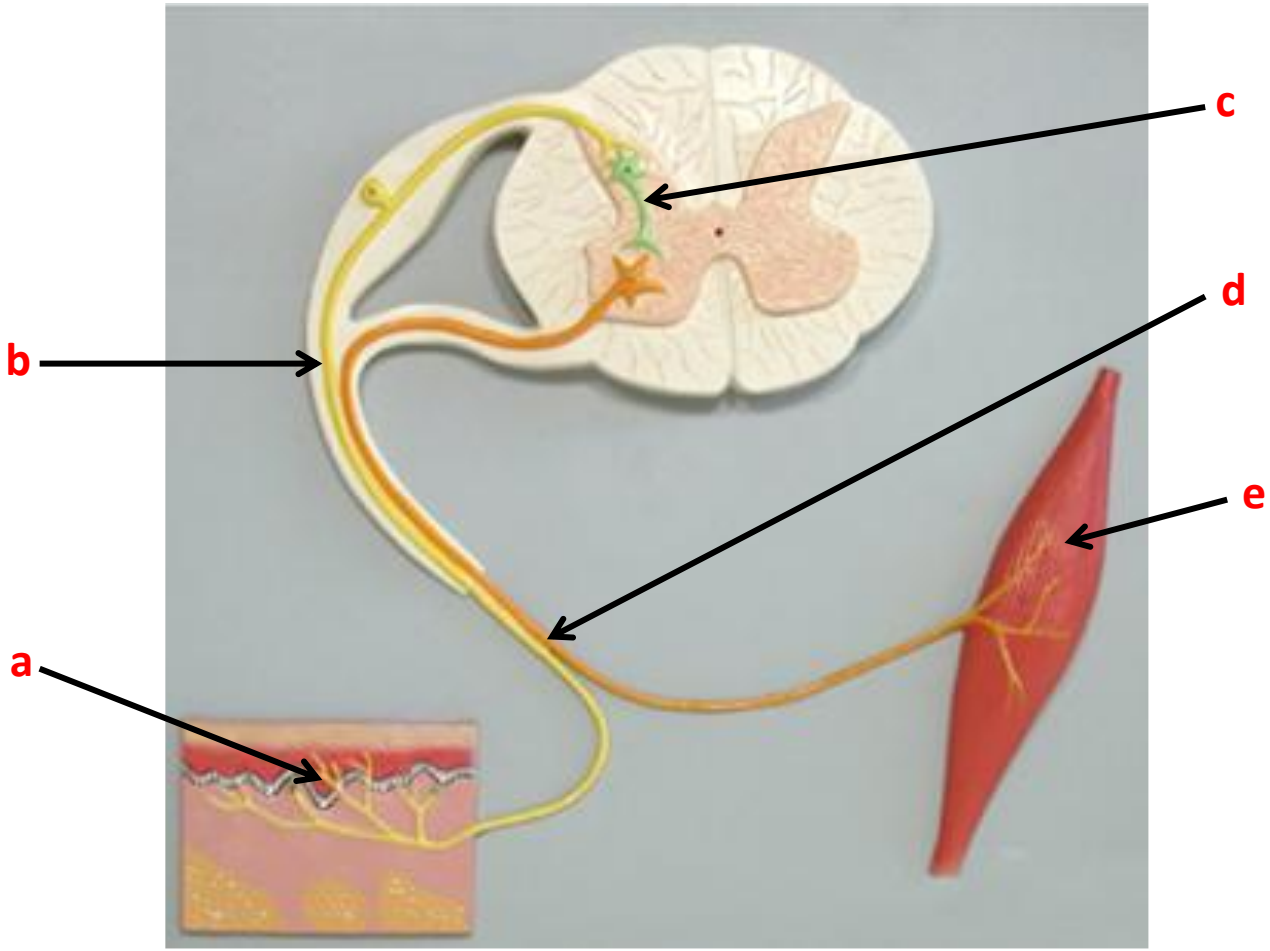
Transverse section of spinal cord

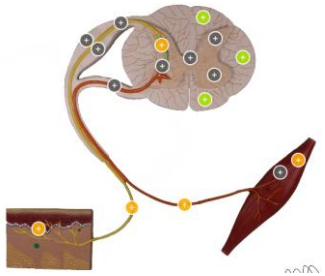


Transverse section of spinal cord

Station 6: The Spinal Reflex Arc

1. Identify and label the structures listed in the appendix on the image below.





Rotational Image Available:
Rotational Images ->
Nervous System -> Reflex Arc

Question:

1. Describe, in your own words, the steps in a reflex arc.

Station 7: Localization of Cutaneous Receptors

Procedure:

1. With your eyes closed, have your lab partner touch your fingertip with either one or two points of the esthesiometer (2 point discriminator). In each case, let your partner know whether it "feels like" one point or two points. Start with the points a large distance apart and gradually bring them closer together until you cannot distinguish between one and two points.

2. Measure in millimeters the distance between the two points of the esthesiometer where it felt like **one point** when applied to the skin. Make each two-point determination three times and average the results. Record your results in Table 1.

3. Repeat the experiment at several locations, on the palm of the hand, dorsum of the hand, ventral and dorsal surfaces of the forearm, back of the neck, shoulder, face (be careful). For each trial, take three readings and average the three. Record your results on Table 1.

Table 1: Distribution of cutaneous receptors

Location	Measurement (mm) Record when it felt like one point. Repeat three times for each region.		Average measurement (mm)
1. Fingertip	1		
	2		
	3		
2. Palm of the hand	1		
	2		
	3		
3. Dorsum of the hand	1		
	2		
	3		
4. Ventral surface of the forearm	1		
	2		
	3		
5. Back of the neck	1		
	2		
	3		
6. Shoulder	1		
	2		
	3		
7. Face	1		
	2		
	3		

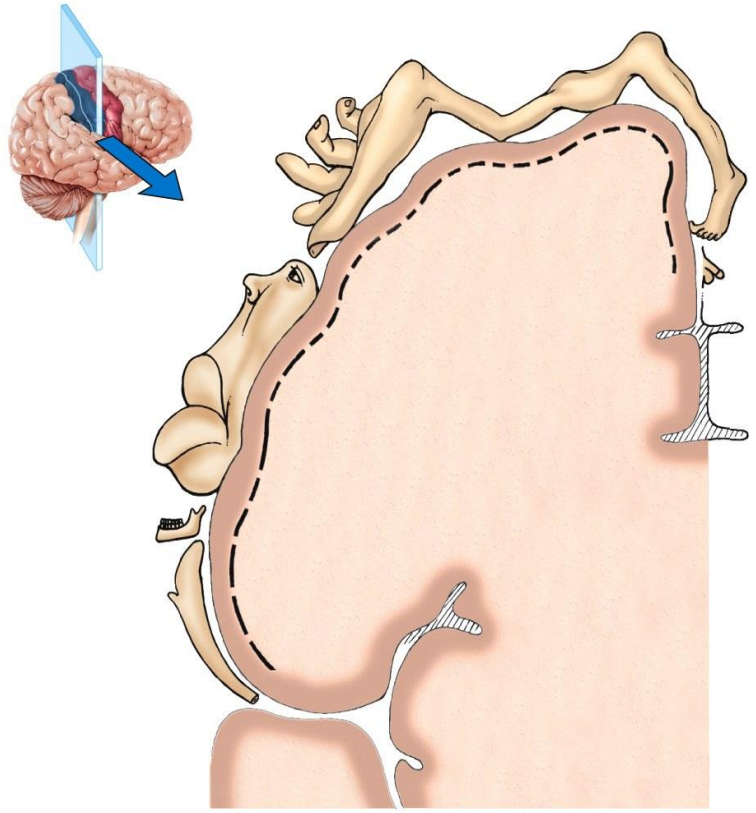
Station 7: Localization of Cutaneous Receptors (continued)

Questions:

1. Which body region from your analysis contained the highest concentration of cutaneous receptors? Can you theorize as to why a certain region would contain more cutaneous receptors than another.

2. What do you conclude about the distribution of sensory receptors in the skin?

3. Compare your conclusion about receptor distribution to the illustration of the homunculus.



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Station 8: Patellar Reflex

Procedure:

1. Have your partner sit with his/her eyes closed, knees flexed, and legs hanging loosely. This stretches the patellar ligament.

2. Feel the tendon between the patella and tibial tuberosity. Gently strike the patellar ligament with the rubber hammer. Perform on both knees.

3. Record your observations in Table 2 under the control condition. This observation should include the speed and vigor of the reflex, and the extent to which the leg kicks out.

4. Now, perform the reflex while the student is concentrating on the following numbers below and trying to perform the addition in his or her head.
   10 + 9 + 4 + 3 + 4 + 7 + 5 + 1 =?.
   Record your observations.

5. Now have the subject concentrate on the reflex itself (this being the movement of the leg kicking outward). Hit the patellar tendon with the hammer and record your observations.

7. Have the student kick out his or her leg when they hear a stimulus. The stimulus can be a snap of the fingers, clap or tap on the table by another student. The student providing the sound should not be seen by the subject performing the voluntary movement (have the volunteer close his/her eyes). Record the observations.

8. Have the student place their hands on the lab bench and attempt to pull the bench upwards while performing the reflex. Record observations.

Table 2: Patellar reflex observations

Action	Observations
Initial tapping (control condition)	
Concentrating on a math problem	
Concentrating on the reflex	
Voluntary leg extension in response to auditory cue	
Pulling bench upwards	

Questions:

1. Under what condition did you observe the fastest reflex? Why?

2. How did the voluntary action compare with the patellar tendon reflex?

3. Explain the patellar tendon reflex using the reflex arc.

Station 9: Plantar Reflex

Procedure:

1. With the subject seated, draw the end of the handle of a reflex hammer up the lateral edge on the sole of the foot. Again, note the action and the speed of the contraction. Record your results below:

Questions:

1. What is the difference between the plantar reflex and the Babinski reflex?

2. What is the difference in the pathway that the patellar reflex and plantar reflex take?

3. What does a positive Babinski sign tell us in the adult?

4. List the muscle and muscle actions for each of the patellar and plantar reflexes.

Station 10: Cranial Nerve Examination

These tests can be tried at home

Table 3: Cranial Nerve Test

Cranial nerve and test. Confirm with lab facilitators if you are not sure of a result	Observation and function of the cranial nerve	Expected Observation
I- Olfactory nerve: Smell and identify the unknown samples and confirm with lab facilitators		Patient should be able to identify each sample
II-Optic nerve: Test acuity using the eye chart. Test colour using the Ishihara plates. Test visual fields by having someone wiggle their fingers in each of the four upper and lower quadrants individually and simultaneously. Perform the pupillary reflex. Use the ophthalmoscope to view your partners fundus.		Patient should be able to read off the eyechart, identify the colours from the plates, see movement of the fingers in the quadrants. The pupil should constrict in both eyes.
III-Oculomotor nerve, IV-Trochlear nerve, VI-Abducens nerve: Have your partner move their fingers in various directions and watch the eye movements. Make sure the head does not move.		The patient should be able to track the movements of the fingers
V-Trigeminal nerve: Test the opthalmic/maxillary and mandibular branches by touching the jaw, cheek and forehead with a light touch and a blunt touch. Gently touch the cornea of your partner and look for the corneal reflex. Ask your partner to open their mouth against resistance. Strike your index finger that is sitting against their chin with a reflex hammer.		The patient should feel the cotton on the various areas and the eye should close upon contact with the cornea. Patient should be able to open their jaw against resistance. The jaw should protrude when struck with a reflex hammer

Station 10: Cranial Nerve Examination (continued)

Table 3: Cranial Nerve Test

Cranial nerve and test. Confirm with lab facilitators if you are not sure of a result	Observation and function of the cranial nerve	Expected Observation
VII-Facial nerve: Ask your partner to show their teeth, crease their forehead and puff out their cheeks.		The movements should be performed with relative ease
VIII-Vestibulocochlear nerve: These can be performed in the hallway where it is quiet. Perform the Rinne test by placing the end of a tuning fork against the mastoid process and then near the ear and asking your partner where it is loudest. When the sound disappears from the mastoid process move it to the ear. Perform Weber's test: Place the tuning fork in the center of the forehead and inquire if it is louder in one ear.		Rinne test- the sound should continue to be heard near the ear once it is no longer detectable by transmission through the bone. Weber's test- the sound should be equal in both ears
IX- Glossopharyngeal nerve: Perform the gag reflex or touch the arches of the pharynx		The gag reflex is normally present
X- Vagus nerve: Ask your partner to say a specific sentence. Observe the movement of the uvula		The words should be clear and the uvula should not deviate
XI-Accessory nerve: Ask your partner to shrug their shoulders and turn their head against resistance		The patient should be able to perform the movements accordingly
XII-Hypoglossal nerve: Ask your partner to stick out their tongue.		There should be no unwanted movements or deviation of the tongue

Station 11: Practice Bellringer

In a Bellringer, you will be given a limited time at each station to identify the structures given. Although you can go through this practice Bellringer at your own pace, it is recommended that you move through the stations as you would in a test situation.

Use the table below to record your answers.

1.		11.	
2.		12.	
3.		13.	
4.		14.	
5.		15.	
6.		16.	
7.		17.	
8.		18.	
9.		19.	
10.		20.	

Appendix

This list includes the terms you are responsible for knowing. Use it to guide you through the lab. Any of these structures may appear on the bellringer, possibly on different models or specimens than those used in the lab.

If a structure has a left and a right, be sure to include that in your answer.

Stations 1: The Neuron and Synapse

Nucleus
Dendrite(s)
Cell body
Node of Ranvier
Axon
Axon terminal
Myelin sheath
Neurilemma
Schwann cell
Synaptic vesicles
Neurotransmitters*
Exocytosis*
Synaptic cleft*

Station 2: The Brain

Cerebrum
Cerebellum
Thalamus
Hypothalamus
Midbrain
Pons
Medulla oblongata
Brainstem
Spinal cord
Pituitary gland*
Corpus callosum
Lateral ventricle
Third ventricle
Fourth ventricle
Grey matter
White matter
Basal nuclei
Longitudinal fissure
Frontal lobe
Occipital lobe
Parietal lobe
Temporal lobe
Central sulcus
Lateral sulcus
Transverse fissure
Parieto-occipital sulcus
Precentral gyrus/primary motor area
Postcentral gyrus/primary sensory (somatosensory) area

Station 3: CSF Flow

Lateral ventricles
Third ventricle
Fourth ventricle
Cerebral aqueduct
Interventricular foramen
Choroid plexus
Arachnoid villus
Dura mater
Arachnoid mater
Pia mater
Subarachnoid space
Subdural space
Dural venous sinus
Epidural space
Cerebral cortex
Skull bone

Station 4: Cranial Nerves

Olfactory nerve
Optic nerve
Oculomotor nerve
Trochlear nerve
Trigeminal nerve
Abducens nerve
Facial nerve
Vestibulocochlear nerve
Glossopharyngeal nerve
Vagus nerve
Accessory nerve
Hypoglossal nerve

Station 5: The Spinal Cord

Dorsal (posterior) column
Ventral (anterior) column
Lateral column
Central canal
Dorsal root
Ventral root
Dorsal root ganglion
Grey matter
White matter

Station 6: The Reflex Arc

Receptor
Afferent neuron
Interneuron
Efferent neuron
Effector

***Term is not found in this lab handout, but you are responsible for this term.**